

Intermediate Macro

Chapter 10: The Facts of Growth

Instructor: Mihyun You

Department of Economics
Georgetown University

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1 Chapter 10: The Facts of Growth

10.2 Growth in Rich Countries since 1950

10.4 Thinking about Growth: A Primer



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Key Tools and Concepts

- Chapter 10 introduces **logarithmic scales** for variable plots.
- Chapter 10 develops an **aggregate production function** with two inputs—labor and capital—and **constant returns to scale**.

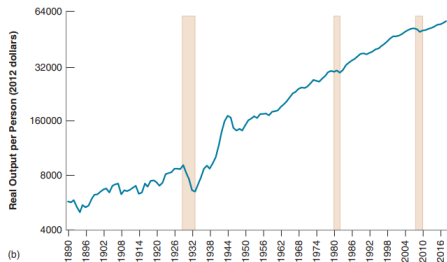
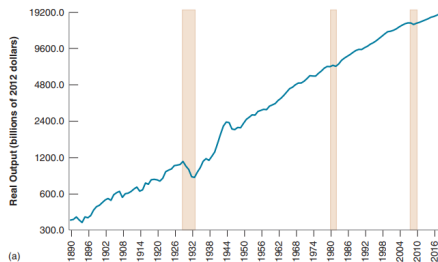
A Famous Quote from Robert Lucas



"Once one starts thinking about growth, it is hard to think about anything else."

US GDP (Total and Per Person) since 1890

10.1 Measuring the Standard of Living



10.1 Measuring the Standard of Living



- The scale on the vertical axis is called a **logarithmic scale**. The defining characteristic of a logarithmic scale is that the same proportional increase in a variable is represented by the same distance on the vertical axis.

Measures of Purchasing Power

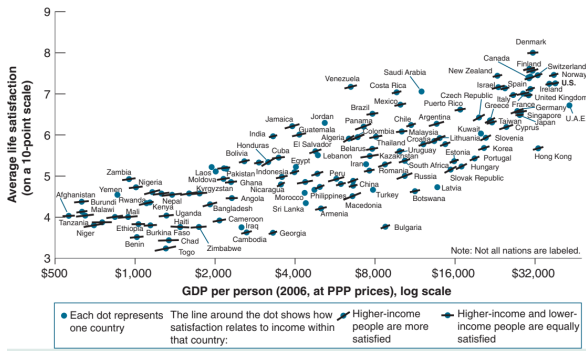
10.1 Measuring the Standard of Living

- The fundamental reason we care about the growth of a country is that we care about the average **standard of living** in the country.
- Output per person provides some measure of a country's standard of living.
- However, to compare **real** output per person across countries, it is important to use a consistent set of prices for the goods produced in each country.
- GDP measures using a common set of prices are called **purchasing power parity (PPP)** numbers.
- PPP numbers also remove exchange rate fluctuations from the calculation of the standard of living.

Does Money Buy Happiness?

10.1 Measuring the Standard of Living

- Output per person is taken as a measure of the standard of living because economists assume that happiness increases with output per person.



The Evolution of Output Per Person in Four Rich Countries since 1950

10.2 Growth in Rich Countries since 1950

- Using PPP numbers, let us examine growth in four large economies—France, Japan, the United Kingdom, and the United States.

Table 10-1 The Evolution of Output per Person in Four Rich Countries since 1950				
	Annual Growth Rate Output per Person (%)	Real Output per Person (2011 dollars)		
	1950–2017	1950	2017	2017/1950
France	2.6	7,025	39,461	5.6
Japan	4.1	2,531	40,374	15.9
United Kingdom	2.1	9,354	39,128	4.2
United States	2.0	14,569	54,995	3.8
Average	2.7	8,370	43,490	5.2
The average in the last line is a simple unweighted average.				
Source: Penn World Table Version 8.1./Feenstra, Robert C., Robert Inklaar and Marcel P. Timmer (2015), "The Next Generation of the Penn World Table" forthcoming American Economic Review.				

The Large Increase in the Standard of Living since 1950

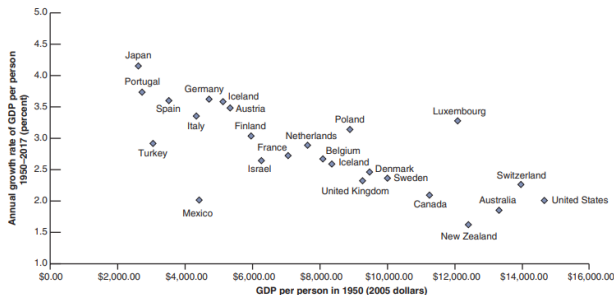
10.2 Growth in Rich Countries Since 1950

- Look at the column on the far right of Table 10-1.
- Output per person has increase by a factor of 3.8 since 1950 in the United States; by a factor of 5.6 in France, and by a factor of 15.9 in Japan.
- These numbers show what is sometimes called the **force of compounding**.
- A better understanding of growth, if it leads to the design of policies that stimulate growth, can have a large effect on the standard of living.

The Convergence of Output per Person

10.2 Growth in Rich Countries Since 1950

- Countries with lower levels of output per person in 1950 have typically grown faster.
- However, not all countries have converged.
- For example, Uruguay and Venezuela were all nearly as rich as France 1950, but far behind by 2017.



Looking across Two Millennia

10.3 A Broader Look across Time and Space

- From a broader historical perspective—say, the past 2000 years or so—growth rates achieved by rich economies since 1950 seem exceptionally high.
- Moreover, the historical record seems more accurately described by leapfrogging than by convergence.
- From the end of the Roman Empire (476-1500), there was essentially no growth of output per person in Europe.
- This period of stagnation of output per person is often called the *Malthusian era*.
- On the scale of human history, therefore, sustained growth of output per person is definitely a recent phenomenon.

Looking across Countries

10.3 A Broader Look across Time and Space

- A closer look at growth since 1960 for a broad sample of 85 countries reveals clear signs of convergence for OECD countries and most Asian economies, but not, in general, for African economies.



Convergence or Divergence?

10.3 A Broader Look across Time and Space

- Many African economies—already poor in 1960—have had negative growth since then.
- In contrast, the Asian countries of Singapore, Taiwan, Hong Kong, and South Korea (sometimes called the **four tigers**) have posted huge gains in average per person output; In 1960, their average output per person was about 15% of that of the United States; by 1970, the ratio had increased to 85%.
- Our main focus will be on the growth in rich and emerging economies.
- There will be some discussion of institutions in slow-growing countries in Chapter 12.

The Aggregate Production Function

10.4 Thinking about Growth: A Primer

- To think seriously about growth, it is necessary to modify the **aggregate production function** to include capital:

$$Y = F(K, N) \quad (10.1)$$

- The function F defines the **state of technology**. The higher the state of technology, the higher $F(K, N)$ for a given K and a given N .

Returns to Scale and Returns to Factors

10.4 Thinking about Growth: A Primer

- The function is assumed to exhibit **constant returns to scale** (CRS) over labor and capital *together*.
- Note that for any number x , a function F has the property of CRS, if

$$F(xK, xN) = xY \quad (10.2)$$

- We have just looked at what happens to production when **both** capital and labor are increased.
- Let us now ask a different question: What should we expect to happen if **only one** of the two inputs in the economy is increased while holding the other is constant?
- Output will increase, but the same increase in one input will lead to smaller and smaller increases in output.

Output per Worker and Capital per Worker

10.4 Thinking about Growth: A Primer

- The CRS assumption also implies that equation (10.1) can be rewritten as

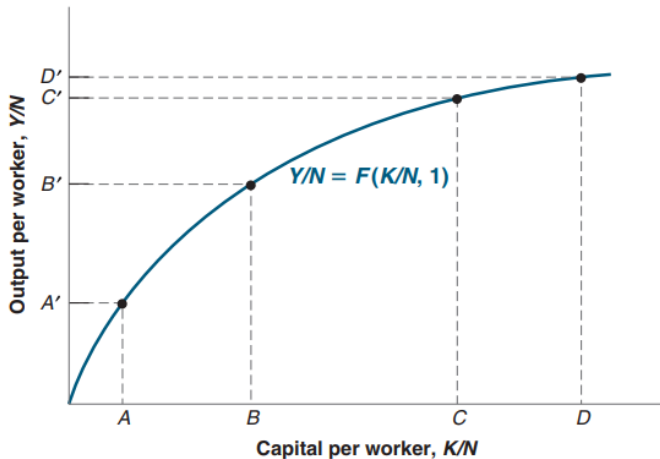
$$\frac{Y}{N} = F\left(\frac{K}{N}, \frac{N}{N}\right) = F\left(\frac{K}{N}, 1\right) \quad (10.3)$$

(+)

- Given CRS, the increase in output per worker $\frac{Y}{N}$ from an extra unit of capital per worker will decline as $\frac{K}{N}$ increases.

Output and Capital per Worker

10.4 Thinking about Growth: A Primer



The Sources of Growth

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- There are two potential sources of growth in output per worker.
 - ① One is **capital accumulation**, and
 - ② The other is **technological progress**—the improvement in the state of technology.

The Sources of Growth

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- There are two potential sources of growth in output per worker.
 - ① One is **capital accumulation**, and
 - ② The other is **technological progress**—the improvement in the state of technology.
- Capital accumulation alone cannot sustain growth indefinitely, because decreasing returns to factor imply that larger and larger increases in capital per worker would be required (higher **saving rate**).
- At some point, society would be unwilling to save the necessary resources to provide for the required increase in capital per worker, and growth would cease.
- Sustained growth requires sustained technological progress.

The Effects of an Improvement in the State of Technology

10.4 Thinking about Growth: A Primer

